Lecture 13 2026-10-13

Today: powered descent guidance

"Convex Programming Approach to Powered Descent Guidance for Mars Landing", B. Açikmese and S. Ploen, Journal of Guidance, Control, & Dynamics, vol. 30, 2007.

$$\mathcal{P} \quad \min_{t_F,\, T_c(t)} \quad \int_0^{t_F} \| T_c(t) \|_2 \, dt$$

subj. to: $\ddot{r}(t) = g + \dfrac{T_c(t)}{m(t)}$ non-convex dynamics

$\dot{m}(t) = -\alpha \| T_c(t) \|_2$ non-convex dynamics

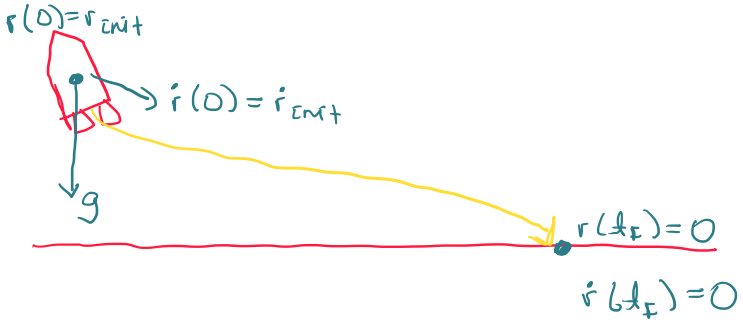
$0 < \rho_1 \leq \| T_c(t) \|_2 \leq \rho_2$ non-convex control

$\| r_d(t) \|_2 \leq \beta r_v(t)$ glideslope constraint

$m(0) = m_{init}$ $\quad m(t_F) \geq 0$

$r(0) = r_{init}$ $\quad r(t_F) = 0$

$\dot{r}(0) = \dot{r}_{init}$ $\quad \dot{r}(t_F) = 0$





ϕ: cant angle

Terms

Terms:

I_{sp}: specific impulse

g_e: gravity

α: mass consumption rate → $\alpha = \frac{1}{I_{sp}\, g \cos\phi}$

n: # of thrusters

T_1, T_2: min & max thrust of each thruster

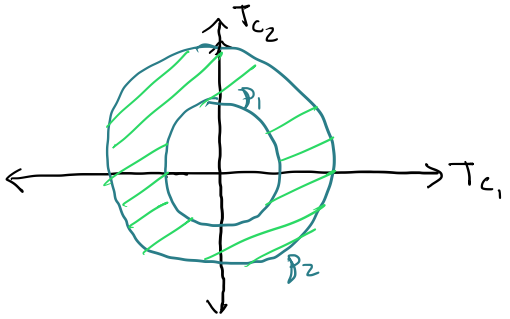
$\rho_1 = n\, T_1 \cos\phi$

$\rho_2 = n\, T_2 \cos\phi$

γ = glideslope angle → $\gamma \in (0, \pi/2)$ → $B = \tan\gamma$

1. Dealing with non-convex control constraints:

$$0 < \rho_1 \leq \|T_c(t)\|_2 \leq \rho_2$$



convexify by introducing slack variable Γ:

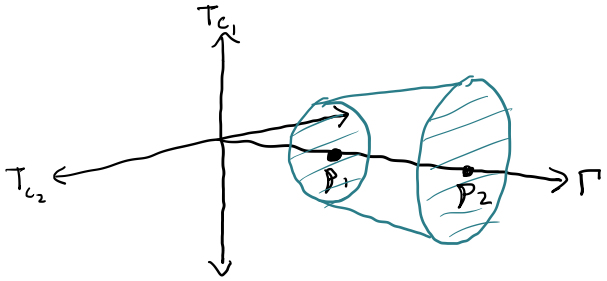
→ re-write as: $\|T_c(t)\|_2 \leq \Gamma$

$$0 < \rho_1 \leq \Gamma \leq \rho_2$$

Can we use this kind of slack variable trick to deal with all kinds of non-convexity?

→ In general, no

But this paper shows this reformulation is lossless i.e., optimal solution of this new problem is optimal w.r.t. original one



So non-convex control constraint is converted:

$$0 < \rho_1 \leq \|T_c(t)\|_2 \leq \rho_2 \implies \begin{array}{l} 0 < \rho_1 \leq \Gamma \leq \rho_2 \\ \|T_c(t)\|_2 \leq \Gamma \end{array}$$

2. Dealing with non-convex dynamics constraints

$$\ddot{r}(t) = g + \frac{T_c(t)}{m(t)} \qquad \dot{m}(t) = -\alpha \|T_c(t)\|_2$$

Let's use a change-of-variables approach:

$$\sigma(t) = \frac{\Gamma(t)}{m(t)} \qquad u(t) = \frac{T_c(t)}{m(t)}$$

↑ scalar term ↑ vector $\in \mathbb{R}^3$

$$\Rightarrow \ddot{r}(t) = g + u(t)$$

$$\dot{m}(t) = -\alpha \|T_c(t)\|_2 \rightarrow \frac{\dot{m}(t)}{m(t)} = -\alpha \underbrace{\frac{\|T_c(t)\|_2}{m(t)}}_{= \frac{\Gamma(t)}{m(t)} = \sigma(t)} = -\alpha\sigma(t)$$

Introduce a new variable: $z(t) := \ln m(t)$

valid to take $\ln(\cdot)$ here since $m(t) > 0$

$$\Rightarrow z(t) = \ln m(t) \rightarrow \dot{z}(t) = \frac{\dot{m}(t)}{m(t)} = -\alpha\sigma(t)$$

$\Updownarrow$

$m(t) = e^{z(t)}$

we also had: $0 < \rho_1 \leq \Gamma(t) \leq \rho_2$

$$\rightarrow \frac{\rho_1}{m(t)} \leq \frac{\Gamma(t)}{m(t)} = \sigma(t) \leq \rho_2$$

$$\rightarrow \frac{\rho_1}{m(t)} \leq \sigma(t) \leq \rho_2$$

Since $z(t) = \ln m(t) \Leftrightarrow m(t) = e^{z(t)} \Leftrightarrow \frac{1}{m(t)} = e^{-z(t)}$

$$\Rightarrow \underbrace{\frac{\rho_1}{m(t)}}_{\rho_1 e^{-z(t)}} \leq \sigma(t) \leq \underbrace{\frac{\rho_2}{m(t)}}_{\rho_2 e^{-z(t)}}$$

p_1



⇒ so we'll use a Taylor series approximation of these constraints

$$e^{-z(t)} \approx 1 + (z(t) - z(t_0)) + \frac{(z(t)-z(t_0))^2}{2!} + \dots$$

$$\Rightarrow \quad e^{-z(t)} \approx e^{-z_0}\left[1 + (z(t) - z(t_0)) + \frac{(z(t)-z(t_0))^2}{2!}\right]$$

$$e^{-z(t)} \approx e^{-z_0}\left[1 + (z(t) - z(t_0))\right]$$

$$\Rightarrow \quad p_1 e^{-z(t)} \approx \underbrace{p_1 e^{-z_0}}_{M_1}\left[1 + (z(t) - z(t_0)) + \frac{(z(t)-z(t_0))^2}{2!}\right]$$

$$p_2 e^{-z(t)} \approx \underbrace{p_2 e^{-z_0}}_{M_2}\left[1 + (z(t) - z(t_0))\right]$$

⇒ Now we have our convex reformulation

$$\min_{T_f, \sigma(t), u(t)} \quad \int_{T_0}^{T_f} \sigma(t)\, dt$$

$$\text{subj. to:} \quad \dot{r}(t) = g + u(t)$$

subj. to: $\ddot{r}(t) = g + u(t)$

$$\dot{z}(t) = -\alpha\sigma(t)$$

$$\|u(t)\|_2 \leq \sigma(t)$$

$$\mu_1\left[1 + (z(t)-z(t_0)) + \frac{(z(t)-z(t_0))^2}{2!}\right] \leq \sigma(t) \leq \mu_2\left[1+(z(t)-z(t_0))\right]$$

$$\underbrace{\ln(m_{inst} - \alpha p_2 t)}_{\mu_1(t)} \leq z(t) \leq \underbrace{\ln(m_{inst} - \alpha p_1 t)}_{\mu_2(t)}$$

$$\|r_d(t)\|_2 \leq \beta r_v(t)$$

$$z_0(0) = m_{inst}$$

$$r(0) = r_{inst} \qquad r(t_F) = 0$$

$$\dot{r}(0) = \dot{r}_{inst} \qquad \dot{r}(t_F) = 0$$

What are the assumptions baked into this model?

1. Deterministic
2. Omits rotational kinematics / dynamics
3. Perfect closed-loop tracking controller
4. Missing drag physics